

Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for full score.

No calculators/computers or books/notes are allowed.

Write your name and student number in all the papers you submit. Papers without name will not be graded.

Do *not* submit your scrap paper, only the paper where the final answers and explanations are written.

Do *not* write your answers with pencil or red/green pen. Only blue/black pen.

Grade = # points/2 + 1

Question 1 (2 pts) Compute the following limits when possible:

a) (1 pts) $\lim_{x \rightarrow 0} \frac{1}{x^3}$

b) (1 pts) $\lim_{x \rightarrow \infty} \frac{5x^3 + x - 10}{2x^2 + 3x + 4}$

Question 2 (3 pts) Let $f(x) = \frac{e^x - 1 - x}{x^2}$. Find $\lim_{x \rightarrow 0} f(x)$.

Hint: When x is close to 0, the exponential function can be written as $e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots$.

Question 3 (3 pts) Show that there exists a solution to the equation $\cos x = x$ in the $[0, 1]$ interval.

Question 4 (3 pts) Let $f(x) = |x|$.

- a) (1 pts) Without proving it formally, explain why the derivative of $f(x)$ doesn't exist at $x = 0$.
- b) (1 pts) Find the derivative of $f(x)$ elsewhere (that is, the function $f'(x)$), expressing it as a piece-wise function and draw its graph.
- c) (1 pts) Find the second derivative of $f(x)$ (that is, the function $f''(x)$), and draw its graph.

Question 5 (3 pts) Consider the following function

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1, \\ 2 & x = 1. \end{cases}$$

Is f a continuous function?

Question 6 (4 pts) Let $g(x, y, z) = x^3 + yz^2$ with

$$x(t) = t^2, \quad y(t) = e^t, \quad z(t) = \sin t.$$

Compute $\frac{d}{dt}g(x(t), y(t), z(t))$.